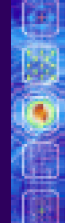
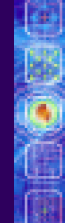




# HUMAN-COMPUTER INTERACTION

THIRD  
EDITION

DIX  
FINLAY  
ABOWD  
BEALE



## chapter 7

### design rules

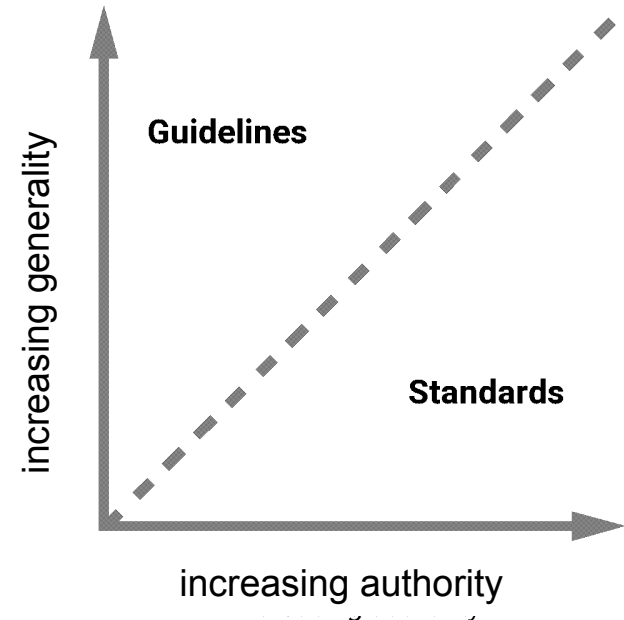
# design rules

Designing for maximum usability  
– the goal of interaction design

- Principles of usability
  - general understanding
- Standards and guidelines
  - direction for design
- Design patterns
  - capture and reuse design knowledge

# types of design rules

- principles
  - abstract design rules
  - low authority
  - high generality
- standards
  - specific design rules
  - high authority
  - limited application
- guidelines
  - lower authority
  - more general application



# Principles to support usability

## Learnability

the ease with which new users can begin effective interaction and achieve maximal performance

## Flexibility

the multiplicity of ways the user and system exchange information

## Robustness

the level of support provided the user in determining successful achievement and assessment of goal-directed behaviour

# Principles of learnability

## Predictability

- determining effect of future actions based on past interaction history
- operation visibility

## Synthesizability

- assessing the effect of past actions
- immediate vs. eventual honesty

# Principles of learnability (ctd)

## Familiarity

- how prior knowledge applies to new system
- guessability; affordance

## Generalizability

- extending specific interaction knowledge to new situations

## Consistency

- likeness in input/output behaviour arising from similar situations or task objectives

# Principles of flexibility

## Dialogue initiative

- freedom from system imposed constraints on input dialogue
- system vs. user pre-emptiveness

## Multithreading

- ability of system to support user interaction for more than one task at a time
- concurrent vs. interleaving; multimodality

## Task migratability

- passing responsibility for task execution between user and system

# Principles of flexibility (ctd)

## Substitutivity

- allowing equivalent values of input and output to be substituted for each other
- representation multiplicity; equal opportunity

## Customizability

- modifiability of the user interface by user (adaptability) or system (adaptivity)



# Principles of robustness

## Observability

- ability of user to evaluate the internal state of the system from its perceivable representation
- browsability; defaults; reachability; persistence; operation visibility

## Recoverability

- ability of user to take corrective action once an error has been recognized
- reachability; forward/backward recovery; commensurate effort

# Principles of robustness (ctd)

## Responsiveness

- how the user perceives the rate of communication with the system
- Stability

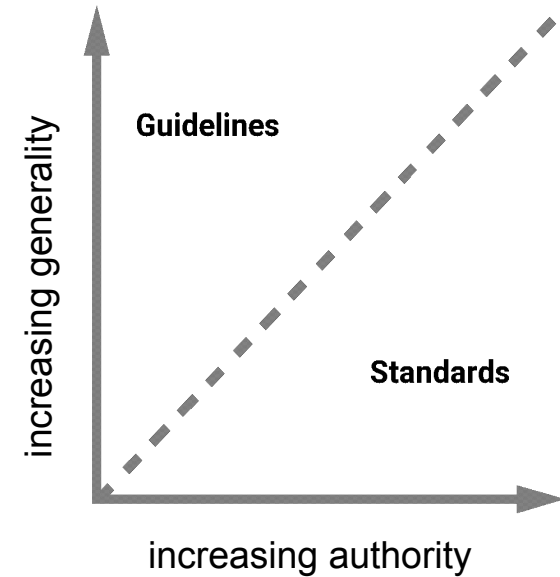
## Task conformance

- degree to which system services support all of the user's tasks
- task completeness; task adequacy

# Using design rules

## Design rules

- suggest how to increase usability
- differ in generality and authority



# Standards

- set by national or international bodies to ensure compliance by a large community of designers standards require sound underlying theory and slowly changing technology
- hardware standards more common than software high authority and low level of detail
- ISO 9241 defines usability as effectiveness, efficiency and satisfaction with which users accomplish tasks

# Guidelines

- more suggestive and general
- many textbooks and reports full of guidelines
- abstract guidelines (principles) applicable during early life cycle activities
- detailed guidelines (style guides) applicable during later life cycle activities
- understanding justification for guidelines aids in resolving conflicts

# Golden rules and heuristics

- “Broad brush” design rules
- Useful check list for good design
- Better design using these than using nothing!
- Different collections e.g.
  - Nielsen’s 10 Heuristics (see Chapter 9)
  - Shneiderman’s 8 Golden Rules
  - Norman’s 7 Principles

# Shneiderman's 8 Golden Rules

- 1. Strive for consistency*
- 2. Enable frequent users to use shortcuts*
- 3. Offer informative feedback*
- 4. Design dialogs to yield closure*
- 5. Offer error prevention and simple error handling*
- 6. Permit easy reversal of actions*
- 7. Support internal locus of control*
- 8. Reduce short-term memory load*

# Norman's 7 Principles

- 1. Use both knowledge in the world and knowledge in the head.*
- 2. Simplify the structure of tasks.*
- 3. Make things visible: bridge the gulfs of Execution and Evaluation.*
- 4. Get the mappings right.*
- 5. Exploit the power of constraints, both natural and artificial.*
- 6. Design for error.*
- 7. When all else fails, standardize.*



# HCI design patterns

- An approach to reusing knowledge about successful design solutions
- Originated in architecture: Alexander
- A pattern is an invariant solution to a recurrent problem within a specific context.
- Examples
  - Light on Two Sides of Every Room (architecture)
  - Go back to a safe place (HCI)
- Patterns do not exist in isolation but are linked to other patterns in *languages* which enable complete designs to be generated

# HCI design patterns (cont.)

- Characteristics of patterns
  - capture design practice not theory
  - capture the essential common properties of good examples of design
  - represent design knowledge at varying levels: social, organisational, conceptual, detailed
  - embody values and can express what is humane in interface design
  - are intuitive and readable and can therefore be used for communication between all stakeholders
  - a pattern language should be generative and assist in the development of complete designs.

# Summary

## Principles for usability

- repeatable design for usability relies on maximizing benefit of one good design by abstracting out the general properties which can direct purposeful design
- The success of designing for usability requires both creative insight (new paradigms) and purposeful principled practice

## Using design rules

- standards and guidelines to direct design activity